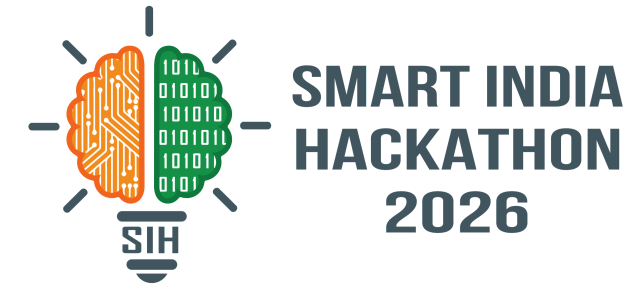


SMART INDIA HACKATHON 2026



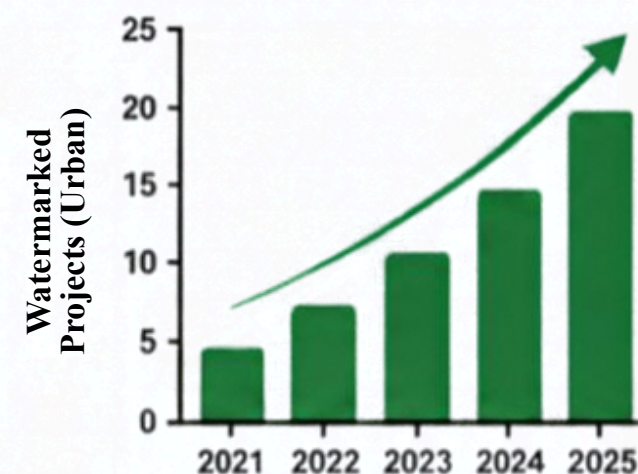
- Problem Statement ID : **SIH26015**
- Problem Statement Title: **Application of Geospatial Techniques for Visualization and Analysis to Interpret Geo-Coded Images to Enhance Watershed Development Outcomes**
- Theme : **Agriculture, FoodTech & Rural Development**
- PS Category : **Software**
- Team ID: **130845**
- Team Name: **Stumble Guys**



Problem Context

-  Geo-tagged field photos are used only for documentation.
-  Satellite data and GIS information are scattered across sources.
-  Difficult to monitor **spatial and temporal changes** in watershed areas.

Why it Matters?



Growing watershed projects require better monitoring solutions.

Innovation & Uniqueness



Integrated geo-photos, satellite data and GIS layers



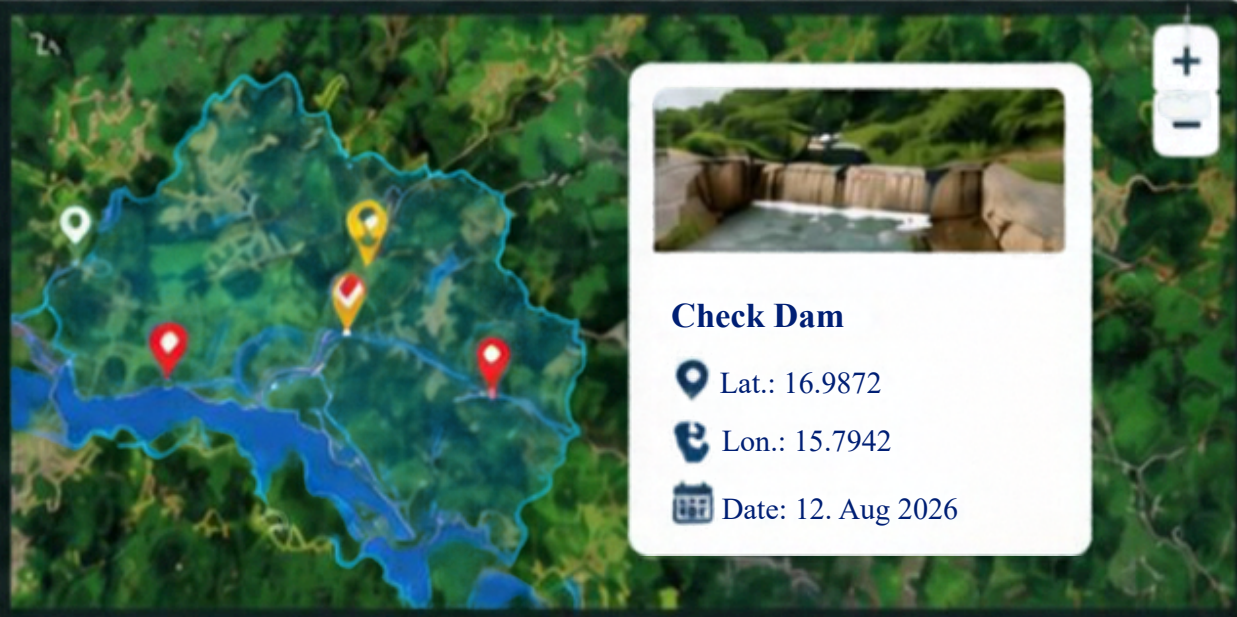
Real-time monitoring and scalable across regions



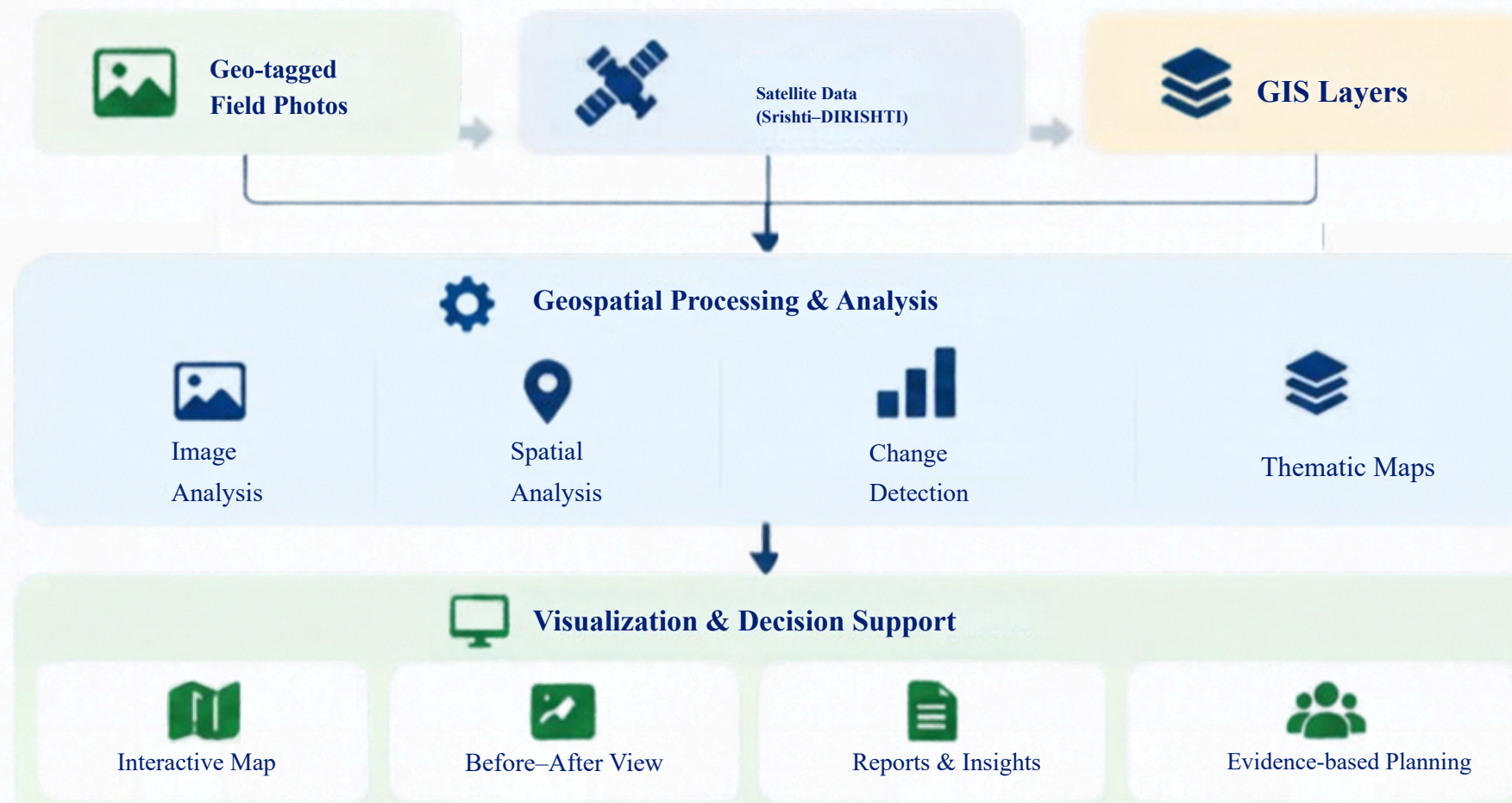
Visual spatial & temporal insights for better decisions

Proposed Solution

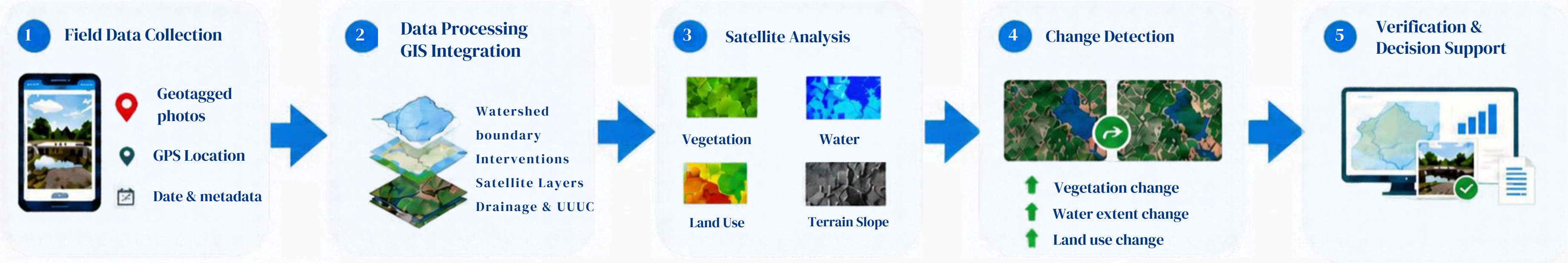
- ☒ Interventions
- ☒ Interventions
- ☒ Geo-tagged Photos
- ☒ Vegetation (NDVI)
- ☒ Water Bodies
- ☒ Drainage Network
- ☒ Land Use



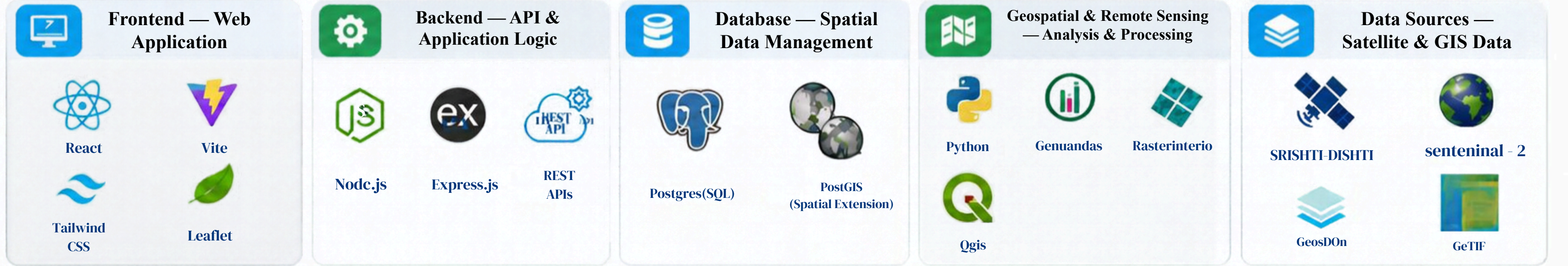
Solution Flow



 **Video :** [Youtube Link](#) [Drive Link](#)
Github Repo: [Repo Link](#)



TECHNOLOGY STACK



IMPLEMENTATION FLOW



Feasibility

Technical Feasibility

- Uses established GIS & Remote Sensing tools
- Integrates geo-tagged field images
- Uses SRISHTI-DRISHTI satellite data
- Supports standard geospatial processing

Operational Feasibility

- User-friendly web-based platform
- Easy data upload and visualization
- Supports planners and field officers
- Applicable across multiple watershed areas

Economic Feasibility

- Primarily software-based solution
- Uses existing geo-photos, satellite & GIS datasets
- Reduces manual data compilation
- Scalable for district/state-level deployment

Viability



Addresses Real Need
Supports watershed monitoring, planning and assessment.



Scalable
Can expand from one watershed → multiple districts → state level.



Replicable
Framework can be adapted to different regions and watershed programs.



Long-Term Adoption
Enables continuous spatial monitoring and supports data-driven watershed planning.

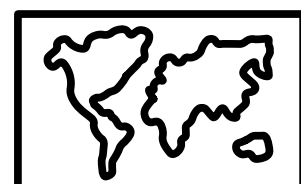
Implementation Roadmap

Data collection and integration



Geo-photos + Satellite Data + GIS Layers

MVP Development



Integrated GIS Web Platform

Pilot Testing



Test on a selected watershed

Scale & Deploy



District → State Level

Key Challenges & Mitigation

Challenge

- Different data formats and coordinate systems
- 30 m resolution may not capture small structures
- Seasonal variation in satellite data

Mitigation Approach

- Standardize projections and validate metadata.
- Use geo-tagged field photos as ground-level evidence.
- Use multi-date imagery and compare appropriate time periods.

IMPACT AND BENEFITS

FOR GOVERNMENT & ADMINISTRATORS



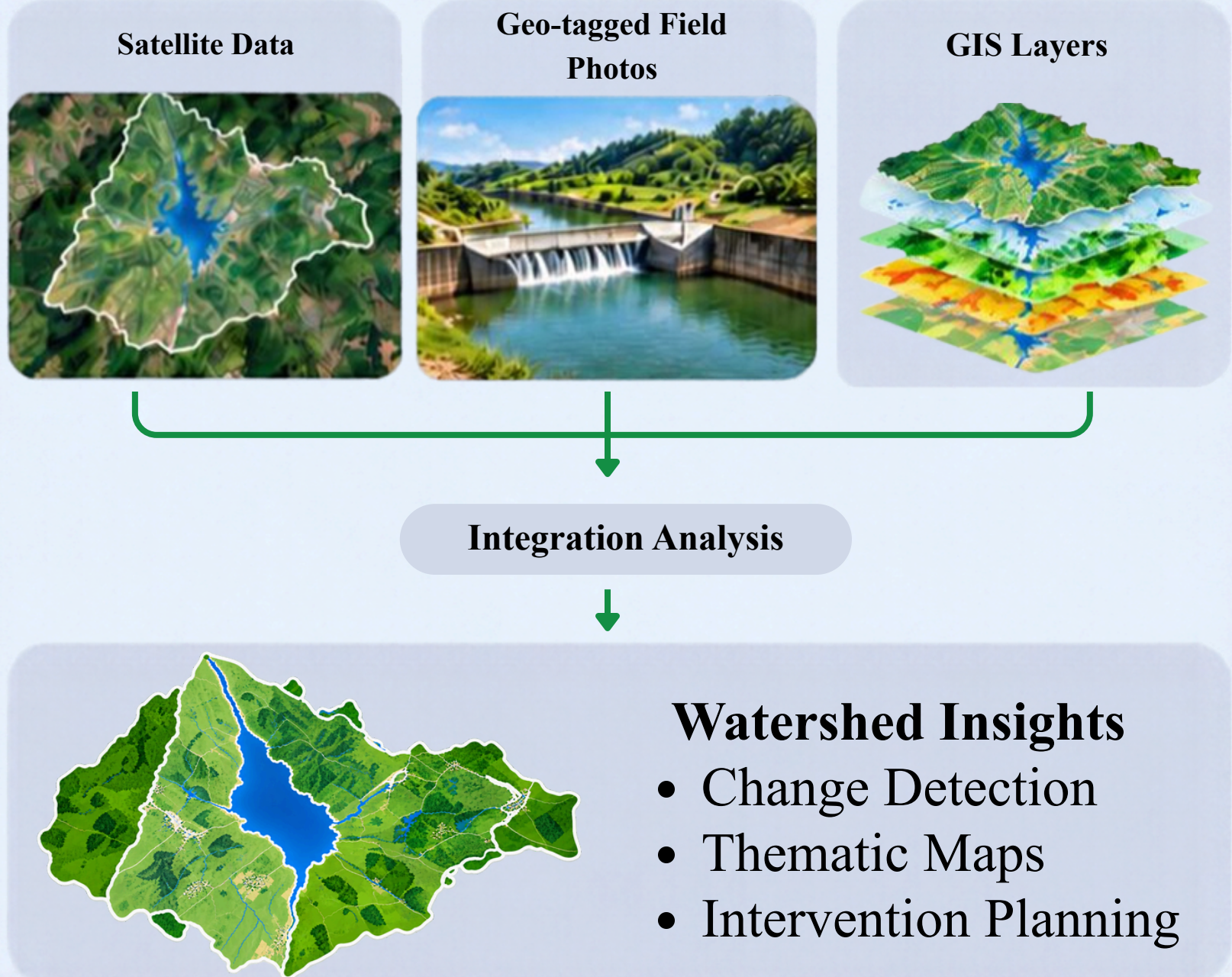
Better Planning
Data-driven decisions for watershed interventions.
Efficient Resource Use



Efficient Resource Use
Helps prioritize critical areas.



Transparent Monitoring
Tracks progress and outcomes effectively.



FOR COMMUNITIES & ENVIRONMENT



Improved Water Security
Supports sustainable water resource management.



Environmental Conservation
Helps maintain soil and vegetation health.



Long-Term Community Benefits
Supports resilient rural communities.

OVERALL IMPACT



Sustainable Watershed Development



Data-Driven Governance



Climate Resilience



Scalable Adoption

References

Bhavan | ISRO–NRSC

Satellite Data

USGS EarthExplorer

Remote Sensing

FAO | Watershed Management

Guidelines

OGGIS Documentation

GIS Tools

NASA SRTM / Earth Data

Elevation Data

Research & Inspiration

Watershed Delineation Studies

Techniques and methodologies for delineating watersheds using DEM data.

Land Use / Land Cover Analysis

Studies on land cover classification and change detection using satellite imagery (Landsat, Sentinel).

Watershed Management Guidelines

Best practices from government and international organizations for sustainable watershed management.